

|                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |                       |                   |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------|-------------------|-----|-----|-----|-----|-----|-----|----------------|------------------------|-------------------|-----------------------|-------------------|-----|-----|------|-----|-------|-----|----------------------------------|------|---|------|-----|---|------|---|------|-----|---|------------|----------|-------|-----|---|------|---|-------|------------|----------|
| <div>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</div> <div>Winter Examination – 2022</div> <div>Course: B. Tech.                      Branch: Civil Engineering                      Semester :V</div> <div>Subject Code &amp; Name: BTHM505 Project Management</div> <div>Max Marks: 60                      Date: 14/02/2023                      Duration: 3 Hr.</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |                       |                   |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |                       |                   |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
|                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (Level/CO)        |                       |                   |     |     |     |     |     |     | Mark<br>s      |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                       |                   |     |     |     |     |     |     | <b>12</b>      |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>1.A)</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Explain Bar chart, its aspects, advantages and dis-advantages.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                   |                       |                   |     |     |     |     |     |     | <b>CO1</b>     | <b>6</b>               |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>1.B)</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Draw the network and calculate TE &amp; TL for all activates for the following</b> <table><tr><td>Activity (i-j)</td><td>1-2</td><td>2-3</td><td>2-4</td><td>3-5</td><td>3-6</td><td>4-5</td><td>4-7</td><td>5-8</td><td>6-8</td><td>7-8</td></tr><tr><td>Duration (t<sup>ij</sup>) Days</td><td>5</td><td>2</td><td>6</td><td>4</td><td>4</td><td>2</td><td>3</td><td>7</td><td>8</td><td>2</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |                       |                   |     |     |     |     |     |     | Activity (i-j) | 1-2                    | 2-3               | 2-4                   | 3-5               | 3-6 | 4-5 | 4-7  | 5-8 | 6-8   | 7-8 | Duration (t <sup>ij</sup> ) Days | 5    | 2 | 6    | 4   | 4 | 2    | 3 | 7    | 8   | 2 | <b>CO1</b> | <b>6</b> |       |     |   |      |   |       |            |          |
| Activity (i-j)                                                                                                                                                                                                                                                                                                                                                                         | 1-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2-3               | 2-4                   | 3-5               | 3-6 | 4-5 | 4-7 | 5-8 | 6-8 | 7-8 |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| Duration (t <sup>ij</sup> ) Days                                                                                                                                                                                                                                                                                                                                                       | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2                 | 6                     | 4                 | 4   | 2   | 3   | 7   | 8   | 2   |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>1.C)</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Prepare table and calculate EST, EFT,LST,LFT, and Total Float associated with the project in Q.no.1(B) , Find critical path &amp; project Duration</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |                       |                   |     |     |     |     |     |     | <b>CO1</b>     | <b>6</b>               |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                             | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                       |                   |     |     |     |     |     |     | <b>12</b>      |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>2.A)</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Explain briefly Direct cost and indirect cost for construction project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                       |                   |     |     |     |     |     |     | <b>CO2</b>     | <b>6</b>               |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| <b>2.B)</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>The following table gives the data for the duration and costs of each activity of project network. The indirect cost of the project 3000Rs/Week.</b> <table><tr><td>Activity</td><td>Normal Duration (Week)</td><td>Normal Cost (Rs.)</td><td>Crash Duration (Week)</td><td>Normal Cost (Rs.)</td></tr><tr><td>1-2</td><td>6</td><td>7000</td><td>3</td><td>14500</td></tr><tr><td>1-3</td><td>8</td><td>4000</td><td>5</td><td>8500</td></tr><tr><td>2-3</td><td>4</td><td>6000</td><td>1</td><td>9000</td></tr><tr><td>2-4</td><td>5</td><td>8000</td><td>3</td><td>15000</td></tr><tr><td>3-4</td><td>5</td><td>5000</td><td>3</td><td>11000</td></tr></table> <b>Draw network. Find project duration and show critical path based on normal duration and corresponding total project cost. Calculate cost slope for each activity.</b> |                   |                       |                   |     |     |     |     |     |     | Activity       | Normal Duration (Week) | Normal Cost (Rs.) | Crash Duration (Week) | Normal Cost (Rs.) | 1-2 | 6   | 7000 | 3   | 14500 | 1-3 | 8                                | 4000 | 5 | 8500 | 2-3 | 4 | 6000 | 1 | 9000 | 2-4 | 5 | 8000       | 3        | 15000 | 3-4 | 5 | 5000 | 3 | 11000 | <b>CO2</b> | <b>6</b> |
| Activity                                                                                                                                                                                                                                                                                                                                                                               | Normal Duration (Week)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Normal Cost (Rs.) | Crash Duration (Week) | Normal Cost (Rs.) |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| 1-2                                                                                                                                                                                                                                                                                                                                                                                    | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 7000              | 3                     | 14500             |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| 1-3                                                                                                                                                                                                                                                                                                                                                                                    | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4000              | 5                     | 8500              |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| 2-3                                                                                                                                                                                                                                                                                                                                                                                    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 6000              | 1                     | 9000              |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| 2-4                                                                                                                                                                                                                                                                                                                                                                                    | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 8000              | 3                     | 15000             |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |
| 3-4                                                                                                                                                                                                                                                                                                                                                                                    | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5000              | 3                     | 11000             |     |     |     |     |     |     |                |                        |                   |                       |                   |     |     |      |     |       |     |                                  |      |   |      |     |   |      |   |      |     |   |            |          |       |     |   |      |   |       |            |          |

| 2.C)           | Determine the optimum duration of project and the corresponding minimum cost associated with the project in Q.no.2 (B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CO2                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----|---|---|----|-----|---|---|----|-----|---|---|----|-----|---|---|----|-----|---|----|----|-----|---|---|----|-----|---|---|---|--|--|
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |                             |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| Q. 3           | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             | 12                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 3.A)           | Differentiate between CPM and PERT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO3                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 3.B)           | Draw the network and find the project duration & Critical path for the following                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CO3                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
|                | <table border="1"> <thead> <tr> <th>Activity (i-j)</th><th><math>t_o</math><br/>(Optimistic Time)</th><th><math>t_L</math><br/>(Most likely Time)</th><th><math>t_p</math><br/>(Pessimistic Time)</th></tr> </thead> <tbody> <tr> <td>1-2</td><td>6</td><td>9</td><td>18</td></tr> <tr> <td>1-3</td><td>5</td><td>8</td><td>17</td></tr> <tr> <td>2-4</td><td>4</td><td>7</td><td>22</td></tr> <tr> <td>3-4</td><td>4</td><td>7</td><td>16</td></tr> <tr> <td>4-5</td><td>4</td><td>10</td><td>22</td></tr> <tr> <td>2-5</td><td>4</td><td>7</td><td>10</td></tr> <tr> <td>3-5</td><td>2</td><td>5</td><td>8</td></tr> </tbody> </table> | Activity (i-j)              | $t_o$<br>(Optimistic Time)  | $t_L$<br>(Most likely Time) | $t_p$<br>(Pessimistic Time) | 1-2 | 6 | 9 | 18 | 1-3 | 5 | 8 | 17 | 2-4 | 4 | 7 | 22 | 3-4 | 4 | 7 | 16 | 4-5 | 4 | 10 | 22 | 2-5 | 4 | 7 | 10 | 3-5 | 2 | 5 | 8 |  |  |
| Activity (i-j) | $t_o$<br>(Optimistic Time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $t_L$<br>(Most likely Time) | $t_p$<br>(Pessimistic Time) |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 1-2            | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 9                           | 18                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 1-3            | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 8                           | 17                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 2-4            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7                           | 22                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 3-4            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7                           | 16                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 4-5            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10                          | 22                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 2-5            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7                           | 10                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 3-5            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5                           | 8                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 3.C)           | Find Standard Deviation and Z value if the project mentioned in Q. no 3 (B) has the completed in 35 Days                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CO3                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |                             |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| Q.4            | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             | 12                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 4.A)           | Explain Demand and supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO4                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 4.B)           | Explain different types of interests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CO4                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 4.C)           | Explain the importance of economics in the construction industry (Civil Engineering field).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO4                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |                             |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| Q. 5           | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             | 12                          |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 5.A)           | Explain the uses, advantages, and limitations of Break-even analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO5                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 5.B)           | Explain the importance of Total Quality Management (TQM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO5                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
| 5.C)           | Explain the uses of computer software (Microsoft Project and Primavera) in project management.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO5                         | 6                           |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |
|                | *** End ***                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |                             |                             |                             |     |   |   |    |     |   |   |    |     |   |   |    |     |   |   |    |     |   |    |    |     |   |   |    |     |   |   |   |  |  |

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